

7. Other Requirements

- Database requirements: the implementation should store verification requests, claim results, evidence/source information, and logs when persistence is enabled.
- Data format requirement: API requests and responses should use a consistent structured format such as JSON.
- Configuration requirement: trusted sources, confidence thresholds, and other configurable parameters should be maintained separately from core verification logic.
- Documentation requirement: the project should document setup, API usage, source configuration, and known limitations.
- Future extensibility: additional trusted sources, knowledge graphs, verification methods, and AI applications should be addable without redesigning the complete system.

Appendix A: Glossary

AI: Artificial Intelligence.

API: Application Programming Interface used for communication between software components.

Claim: A factual statement extracted from AI-generated text that can be checked against evidence.

Dehallucination: The process of detecting and reducing unsupported or incorrect statements produced by an AI system.

Evidence: Information retrieved from a trusted source and used to verify a claim.

Fact Verification: Comparison of a factual claim with available evidence to determine its support.

Hallucination: AI-generated information that is unsupported, incorrect, or presented as factual without adequate basis.

Knowledge Graph: A structured representation of entities and relationships that can be used as a source of factual information.

Confidence Score: A value or level indicating how strongly the system supports a verification result.

Trusted Source: A configured information source considered suitable for evidence retrieval.

Verification Result: The status and supporting information produced after comparing a claim with evidence.

Reviewer: An authorized user who examines flagged or uncertain verification results.

Appendix B: Field Layouts

The following fields represent the main data items expected in the verification request/result and can be used as the basis for the required field-layout spreadsheet in the project.

Field	Data Type	Length	Mandatory	Description
Request ID	String	TBD	Yes	Unique identifier for a verification request.

Generated Response	Text	TBD	Yes	AI-generated text submitted for verification.
Claim ID	String	TBD	No	Identifier for an extracted claim.
Claim Text	Text	TBD	No	Factual claim extracted from the response.
Verification Status	String	30	Yes	Supported, Contradicted, or Uncertain.
Confidence	Numeric	TBD	No	Confidence value or level assigned to the result.
Evidence	Text	TBD	No	Relevant information used to verify the claim.
Source	String/URL	TBD	No	Trusted source associated with the evidence.
Timestamp	Date/Time	TBD	Yes	Time at which the verification was performed.
User/Client ID	String	TBD	No	Identifier of the requesting user/application where applicable.
Error Code	String	TBD	No	Code describing a validation or retrieval failure.

Report requirements: A verification report should contain the request ID, original response, extracted claims, verification status, confidence information, supporting evidence, sources, timestamp, and any errors or warnings.

Appendix C: Requirement Traceability Matrix

The traceability matrix links the major functional requirements to the system features and the corresponding analysis/use-case areas. It can be expanded during implementation and testing.

Requirement Range	System Feature	Related Use Case / Model
FR-01 – FR-04	Response Input and Validation	Submit AI generated response
FR-05 – FR-07	Claim Extraction	Verify response / Sequence Diagram

FR-08 – FR-11	Trusted Evidence Retrieval	Check with trusted sources / Sequence Diagram
FR-12 – FR-15	Fact Verification	Verify response / Sequence Diagram
FR-16 – FR-18	Confidence Scoring	View verification result
FR-19 – FR-22	Hallucination Detection and Response Processing	Verify response / View result
FR-23 – FR-25	Evidence and Result Presentation	View verification result / View supporting evidence
FR-26 – FR-28	Verification API	Use verification API
FR-29 – FR-30	Trusted Source Management	Manage trusted sources

Note: Detailed test-case traceability will be added during the implementation/testing phase. The PES guideline specifies that later testing should cover the implemented use cases with test cases.